

C895

Astronomy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Using a small refractor like one on display at a local history museum, an observer sees Venus pass through a full set of phases, including gibbous, while its apparent size changes greatly. The telescope magnifies 20 times. What does this show?

- A. Only a 20-power telescope can reveal planetary orbits.
- B. Venus orbits Earth between Earth and the Sun.
- C. Venus orbits the Sun.
- D. Earth's shadow falls across Venus.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

At a star party hosted by an amateur club with 60 members, the Moon is full. The club's telescope is on loan from a nearby college. About when did the full Moon rise?

- A. Around noon
- B. Around sunset
- C. Around sunrise
- D. Around midnight

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A student claims Moon phases happen because Earth's shadow covers part of the Moon. Which observation best contradicts this?

- A. Lunar eclipses happen every month at new Moon.
- B. A thin crescent Moon appears close to the Sun in the sky, where Earth's shadow cannot reach.
- C. The Moon always shows the same face to Earth.
- D. The Moon makes its own light at full phase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Earth is closest to the Sun in early January, about 147 million km away, versus about 152 million km in July. A science center with a new planetarium shows these figures. Why is it winter in the Northern Hemisphere in January?

- A. The whole Earth has winter in July, when it is farthest away.
- B. Earth is farthest from the Sun in January.
- C. The Sun gives off less energy in winter.
- D. The Northern Hemisphere is tilted away from the Sun, so sunlight is lower and days are shorter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

If Earth's axis had no tilt at all, what would happen to seasons?

- A. Seasons would stay the same, because distance causes them.
- B. Every place would have constant summer.
- C. Seasons would become stronger.
- D. Seasons would nearly vanish, with about 12-hour days everywhere all year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A star photographed by a college observatory that opened in 1962 is imagined moved to 3 times its current distance. The star's surface is about 6,000 K. How does its apparent brightness change?

- A. It becomes 9 times brighter.
- B. It becomes 1/3 as bright.
- C. It becomes 1/9 as bright.
- D. It stays the same, because the star's luminosity is unchanged.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Stars A and B have the same luminosity. Star A appears 16 times brighter than star B. How do their distances compare?

- A.** They must be the same distance, since luminosity is equal.
- B.** B is 4 times farther than A.
- C.** A is 4 times farther than B.
- D.** B is 16 times farther than A.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A planetary scientist, whose team uses a supercomputer bought for \$3 million, models the young solar system. Why did the giant planets form farther from the Sun?

- A.** Beyond the frost line, ices condensed, giving more solid material for big cores that pulled in gas.
- B.** The giant planets formed from a separate cloud and were captured later.
- C.** The Sun's gravity pulled heavy material inward and left light gas outside.
- D.** Planets always grow larger the farther they are from their star.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A star of 1 solar mass, like the Sun, which a museum exhibit hall shows as a 3-m sphere, runs out of core fuel. What remnant will it leave?

- A.** A white dwarf
- B.** A neutron star
- C.** A black hole
- D.** It will remain a red giant forever.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Astronomers find a neutron star. What can they infer about the star it came from?

- A.** It was one of the most massive stars known, which always leave neutron stars.
- B.** It formed when a white dwarf slowly cooled.
- C.** It was a low-mass star like the Sun.
- D.** It was a massive star (roughly 8 or more solar masses) that exploded as a supernova.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A nebula 1,500 light-years away was featured in a popular nature documentary. When we view it tonight, what are we seeing?

- A.** The nebula as it was about 1,500 years ago
- B.** The nebula as it will be in 1,500 years
- C.** A nebula that is exactly 1,500 years old
- D.** The nebula as it is right now

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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12.

A galaxy cataloged by a survey run from a mountaintop observatory in Chile lies 200 megaparsecs away. Using $H_0 = 70 \text{ km/s per megaparsec}$, what is its recession speed?

- A.** Zero, because only galaxies near the center of the universe move
- B.** 70 km/s
- C.** 14,000 km/s
- D.** About 2.9 km/s

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

Using a small refractor ~~like one on display at a local history museum~~, an observer sees Venus pass through a full set of phases, including gibbous, while its apparent size changes greatly. ~~The telescope magnifies 20 times.~~ What does this show?

Target word: what the phases prove **Decides:** Gibbous and full phases need Venus to go behind the Sun.

A	Camouflage	Magnification is a side detail; the phase pattern decides it.
B	Confusion set	That geocentric setup would show only crescent phases.
C	Correct	Only a Sun-centered orbit lets us see nearly all of its lit side.
D	Adjacent category	Earth's shadow does not reach Venus.

2. Answer B Concept 2

At a star party ~~hosted by an amateur club with 60 members~~, the Moon is full. ~~The club's telescope is on loan from a nearby college.~~ About when did the full Moon rise?

Target word: rise time **Decides:** Full Moon is opposite the Sun.

A	Adjacent category	First-quarter Moon rises near noon.
B	Correct	Opposite the Sun, it rises as the Sun sets.
C	Inverted operation	That is the new Moon, which sits near the Sun.
D	Confusion set	Third-quarter Moon rises near midnight.

3. Answer B Concept 2 · transfer

A student claims Moon phases happen because Earth's shadow covers part of the Moon. Which observation best contradicts this?

Target word: evidence against the shadow model **Decides:** Earth's shadow points away from the Sun.

A	Inverted operation	Eclipses are rare and happen at full Moon.
B	Correct	The shadow lies opposite the Sun; it reaches the Moon only during a lunar eclipse.
C	Adjacent category	True, but it is about rotation, not phases.
D	Confusion set	The Moon only reflects sunlight.

4. Answer D Concept 3

Earth is closest to the Sun in early January, about 147 million km away, versus about 152 million km in July. ~~A science center with a new planetarium shows these figures.~~ Why is it winter in the Northern Hemisphere in January?

Target word: cause of seasons **Decides:** Axial tilt sets sun angle and day length.

A	Absolute language	Hemispheres have opposite seasons.
B	Inverted operation	The stem gives the opposite: closest in January.
C	Adjacent category	The Sun's output is nearly constant through the year.
D	Correct	Tilt, not distance, drives the seasons.

5. Answer D Concept 3 · transfer

If Earth's axis had no tilt at all, what would happen to seasons?

Target word: seasons without tilt **Decides:** Tilt is the cause.

A	Confusion set	Distance plays only a minor role.
B	Absolute language	Polar regions would stay cold; there is just no yearly cycle.
C	Inverted operation	Removing the cause weakens the effect.
D	Correct	Only the small distance change would remain.

6. Answer C Concept 4

A star ~~photographed by a college observatory that opened in 1962~~ is imagined moved to 3 times its current distance. ~~The star's surface is about 6,000 K.~~ How does its apparent brightness change?

Target word: new brightness **Decides:** Brightness $\propto 1/d^2$.

A	Inverted operation	Farther means dimmer.
B	Dropped element	Drops the square.
C	Correct	$1/3^2 = 1/9$.
D	Confusion set	Luminosity is fixed; apparent brightness depends on distance.

7. Answer B Concept 4 · transfer

Stars A and B have the same luminosity. Star A appears 16 times brighter than star B. How do their distances compare?

Target word: solve for distance **Decides:** Brightness ratio $16 = (d_B/d_A)^2$.

A	Absolute language	Equal luminosity with unequal brightness means unequal distance.
B	Correct	$\sqrt{16} = 4$.
C	Inverted operation	The brighter star of equal luminosity is closer.
D	Dropped element	Forgets to take the square root.

8. Answer A Concept 5

A planetary scientist, ~~whose team uses a supercomputer bought for \$3 million,~~ models the young solar system. Why did the giant planets form farther from the Sun?

Target word: cause of the planet split **Decides:** The frost line.

A	Correct	Larger cores could capture hydrogen and helium.
B	Adjacent category	All planets formed from the same disk.
C	Confusion set	Temperature, not sorting by gravity, set what could condense.
D	Absolute language	Uranus and Neptune are smaller than Jupiter and Saturn despite being farther.

9. Answer A Concept 6

A star of 1 solar mass, like the Sun, ~~which a museum exhibit hall shows as a 3-m sphere,~~ runs out of core fuel. What remnant will it leave?

Target word: remnant type **Decides:** Low mass.

A	Correct	It sheds its outer layers as a planetary nebula; the core remains.
B	Confusion set	Requires a massive star and a supernova.
C	Absolute language	Not every star becomes a black hole; only the most massive do.
D	Dropped element	Stops at a middle stage; the giant phase ends.

10. Answer D Concept 6 • transfer

Astronomers find a neutron star. What can they infer about the star it came from?

Target word: infer original mass **Decides:** Neutron stars come from massive stars that went supernova.

A	Adjacent category	The most massive stars tend to leave black holes.
B	Confusion set	Cooling white dwarfs become cold white dwarfs, not neutron stars.
C	Inverted operation	Low-mass stars leave white dwarfs.
D	Correct	Only massive cores collapse to neutron density.

11. Answer A Concept 7

A nebula 1,500 light-years away ~~was featured in a popular nature documentary.~~ When we view it tonight, what are we seeing?

Target word: lookback time **Decides:** Light takes 1,500 years to arrive.

A	Correct	The light left it 1,500 years ago.
B	Inverted operation	Light carries the past, not the future.
C	Confusion set	A light-year measures distance, not the object's age.
D	Dropped element	Drops the light's travel time.

12. Answer C Concept 8

A galaxy ~~cataloged by a survey run from a mountaintop observatory in Chile~~ lies 200 megaparsecs away. Using $H_0 = 70$ km/s per megaparsec, what is its recession speed?

Target word: recession speed **Decides:** $v = H_0 \times d$.

A	Absolute language	Expansion has no center; all distant galaxies recede.
B	Wrong stem number	Reports the constant, not the speed.
C	Correct	70×200 .
D	Inverted operation	Divides 200 by 70.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Evidence for a Sun-centered system	Venus showing a full set of phases, including gibbous, means it orbits the Sun; an Earth-centered model allows only crescents.	Thinking phases alone prove nothing about orbits, or blaming Earth's shadow.	Telescope observations of Venus's phases and changing size.
2	Cause and timing of Moon phases	Phases come from how much of the Moon's sunlit half we see; a full Moon is opposite the Sun and rises near sunset.	Earth's shadow causes phases.	A phase plus "when does it rise?" or "why does it look this way?"
3	Cause of seasons	Seasons come from the 23.5° axial tilt, which changes sun angle and day length; Earth is actually closest to the Sun in January.	Distance from the Sun causes seasons.	Perihelion in northern winter, or a no-tilt thought experiment.
4	Inverse-square law for brightness	Apparent brightness $\propto$ luminosity $\div$ distance ² ; triple the distance, one-ninth the brightness.	Dropping the square, or confusing brightness with luminosity.	A distance multiplied or a brightness ratio given.
5	Why terrestrial and giant planets differ	Beyond the frost line, ices condensed, giving more solid material for large cores that captured hydrogen and helium gas.	The Sun's gravity pulled heavy material inward.	Why rocky planets are inside and giants outside.
6	Stellar remnants depend on mass	Low- and Sun-mass stars leave white dwarfs; massive stars explode and leave neutron stars; the most massive leave black holes.	All stars become black holes or supernovae.	A star's mass in solar masses.
7	Light-years and lookback time	A light-year is a distance; an object N light-years away is seen as it was N years ago.	Treating a light-year as a unit of time or seeing distant objects as they are now.	"What are we seeing when we look at... light-years away?"
8	Hubble's law	Recession speed $v = H_0 \times d$ ($H_0 \approx 70$ km/s per megaparsec); every galaxy sees others receding, so there is no center.	Dividing by H_0 , or placing the Milky Way at the center.	A distance in megaparsecs and H_0 .

Course scope (public description): Astronomy provides graduate students seeking initial licensure or endorsement and/or to earn their MA degree in secondary or middle grade science education with essential knowledge of astronomy. This course explores Western history and basic physics of astronomy; phases of the moon and seasons; composition and properties of solar system bodies; stellar evolution and remnants; properties and scale of objects and distances within the universe; and introductory cosmology. A prerequisite for this course is General Physics.